Alvin Reji

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EDUCATION

Northeastern University, Boston, MA **Sep 2024 – Dec 2027**
Candidate for Bachelor of Science in Mechanical Engineering with an Aerospace Minor *GPA: 3.87*
Activities: AerospaceNU, ASME, Materials Engineering Research, TAMID, InterVarsity
Relevant Courses: Statics, Mechanics of Materials, Dynamics, Propulsion, Mechanical Design, Thermodynamics, Fluid Mechanics, Heat Transfer, Computational Fluid Dynamics, Systems Analysis & Control, Electrical Engineering, Mat Science, Engineering Computations

SKILLS

Hardware: Microcontrollers, CNC Routing, 3D Printing, TIG Welding, Circuit Design, Soldering, Laser Cutting, Oscilloscope, ShopBot
Software: SolidWorks, Ansys, Minitab, MS Office, GSuite, Altium, MATLAB, AutoCAD, Python, C++, Creo, MATLAB, MagicDraw
Interests: Rock climbing, Weightlifting, Volleyball, Spikeball, Cycling, Running, Hiking, Billiards, Photography, Videography, Pottery

EXPERIENCE

Massachusetts Institute of Technology **Cambridge, MA**
Teaching Assistant *Aug 2025 – Present*
- Educated 24 students on the iterative design process while introducing Six Sigma quality concepts to improve process reliability
- Fostered a collaborative, inclusive classroom by coaching conflict resolution, and promoting effective decision-making strategies
- Led student teams through design projects by providing structured guidance on engineering problem-solving and teamwork

BAE Systems **Nashua, NH**
Mechanical Engineering Intern *June 2026 – Aug 2026*
- Developed Block Definition and Use Case Diagrams in MagicDraw to define system integration and user-product interactions
- Designed locking mechanisms and cable management solutions for complex systems within classified laboratory environments
- Executed root-cause analyses by developing Linux-based data logging tools and utilizing statistical analysis to determine factors
- Led hardware development for multi-point RF geo-locator system, whilst collaborating cross-functionally with engineering teams
- Designed and fabricated IP53-rated, dynamic antenna modules using Creo and FDM printing to identify and locate distress signals

Johnson & Johnson Medtech (Abiomed) **Danvers, MA**
Product Development Engineering Co-Op *Jan 2026 – June 2026*
- Designed and iterated custom Solidworks components for Instron testing using FDM/SLA printing for product verification tests
- Conducted 100+ mechanical tests simulating clinical-use conditions and optimized test method to improve variability by 73%
- Performed statistical analyses using Minitab to execute TMVs and Gage R&R studies for data-driven design improvements
- Developed a Python/OpenCV pipeline to automate analyses across 30+ experimental iterations, improving design efficiency
- Authored engineering documentation, test protocols, and work instructions ensuring compliancy, traceability, and consistency

CubeSAT at Northeastern University **Boston, MA**
Mechanical Engineer *Sep 2024 – Dec 2025*
- Designed optical component casings for 1U CubeSAT, ensuring structural integrity during launch using FEA and DFMA principles
- Generated ASME Y14.5-compliant engineering drawings using GD&T principles for high tolerance fabrication and assembly

AerospaceNU Rocket Team at Northeastern University **Boston, MA**
Avionics Engineer *Sep 2024 – Dec 2025*
- Developed software for reliable sensor data acquisition on flight computers by implementing signal processing and noise reduction
- Designed flight computer PCBs optimized for power distribution, EMI compliance, and thermal management using KiCAD

University Nanosatellite Program at University at Buffalo **Buffalo, NY**
Guidance, Navigation, & Control Engineer *Aug 2023 – May 2024*
- Developed MATLAB simulations for satellite missions, modeling orbital mechanics, attitude dynamics, and control responses
- Implemented Monte Carlo algorithms and Kalman filters to improve satellite state determination and reduce signal noise by 54%
- Presented mission architecture, GNC simulation results, and subsystem performance analyses during CDR/PDR to AFRL experts

PROJECTS

GE1501 Cornerstone of Engineering at Northeastern University **Boston, MA**
Horse Racing Tracking Collar *Oct 2024 – Dec 2024*
- Designed and fabricated an ESP32-based collar using SolidWorks and FDM 3D printing, ensuring sensor stability and durability
- Implemented GPS and accelerometer integration, enabling accurate horse position and motion tracking with ±1 m positional error
- Developed embedded C++ software for real-time Wi-Fi data transmission and data visualization through a custom PyQt5 GUI

NASA Techrise Student Challenge **Elmont, NY**
Atmospheric Measurement of Noxious Gases Using Sensors (AMONGUS) *Nov 2021 – July 2022*
- Conducted pre-launch testing in low-pressure chambers, improving payload reliability by 30% under high altitude conditions
- Fabricated and assembled payload components using CNC machining and laser cutting, ensuring structural integrity during flight
- Performed system integration, calibration, and validation of environmental sensors, ensuring accurate flight data collection